Akoizo Original Practice - Form 3

Division B | Botany | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) The Casparian strip particularly restricts…

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

A. Unregulated apoplastic entry

B. All photosynthesis

C. Every phloem tube

D. All root growth

Answer: _____

2. (2 points) Ferns reproduce through…

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

A. Spores with a gametophyte stage

B. Flowers and fruit universally

C. Only seeds

D. No meiosis

Answer: _____

3. (2 points) A fungal structure on a lesion is an example of…

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

A. A sign

B. Only a symptom

C. A root system

D. A light-response plateau

Answer: _____

4. (2 points) Do all plant cells have many chloroplasts?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

A. Only every dead xylem cell

B. No

C. Yes

D. Only every root cell

Answer: _____

5. (2 points) Pollination and fertilization are…

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

A. Both meiosis

B. Different events

C. Synonyms

D. Both always seed dispersal

Answer: _____

6. (2 points) Oldest known fossil equals exact origin time?

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

A. Only for pollen

B. Not necessarily

C. Always

D. Only if the fossil is large

Answer: _____

7. (2 points) A herbarium primarily preserves…

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

A. Every soil variable

B. A full evolutionary tree automatically

C. Specimens and collection evidence

D. A continuous gas-exchange record

Answer: _____

8. (2 points) Hormone effects depend on…

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

A. Only leaf color

B. No concentration effects

C. Tissue and context

D. Only the hormone name

Answer: _____

9. (4 points) Explain “Chloroplast” in your own words.

Unit: Photosynthesis and plant energy.
Worked-case data: Low light: 5 bubbles per minute. Medium light: 10 bubbles per minute. High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.
Separate investigation observations:
Low light: 5 bubbles per minute.
Medium light: 10 bubbles per minute.
High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.

Answer: _____
_____
_____

10. (2 points) Vegetative propagation commonly helps preserve…

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

A. No traits

B. Every environmental effect

C. A selected genotype

D. Maximum new genetic variation necessarily

Answer: _____

11. (4 points) Compare apoplast and symplast.

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

Answer: _____
_____
_____

12. (4 points) Name two major seed-plant innovations.

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

Answer: _____
_____
_____

13. (4 points) Distinguish symptom and sign.

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

Answer: _____
_____
_____

14. (4 points) What is an apical meristem’s role?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

Answer: _____
_____
_____

15. (4 points) What commonly forms triploid endosperm?

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

Answer: _____
_____
_____

16. (4 points) Why can older leaves show deficiency first?

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

Answer: _____
_____
_____

17. (4 points) Why are lignified xylem walls useful?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

Answer: _____
_____
_____

18. (4 points) Why is a breeding method not an outcome guarantee?

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

Answer: _____
_____
_____